

ML Week 9

Number of participants: 31



1. The support vector classifier:

9 correct answers
out of 20 respondents



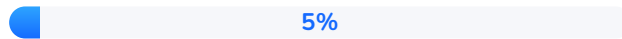
Is more robust than the maximal margin classifier since it allows more support vectors to contribute to the solution.



80%

16 votes

Has no tuning parameters.



5%

1 vote

Cannot overfit since it is linear.



15%

3 votes

Has a closed form solution.



35%

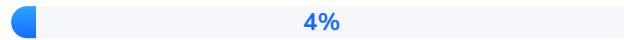
7 votes



2. The support vector machine:

10 correct answers
out of 24 respondents

SVMs model the posterior class probabilities.



1 vote



The SVM is simply a support vector classifier in an extended feature space.



18 votes



The overall budget is also a tuning parameter of SVM.



24 votes



Different kernels result in different classes of prediction functions.



15 votes



3. Which type of features can support vector machine handle without preprocessing?

8 correct answers
out of 23 respondents

Numerical and categorical



9 votes



Only numerical



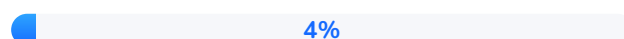
9 votes

Only categorical



5 votes

Neither numerical nor categorical



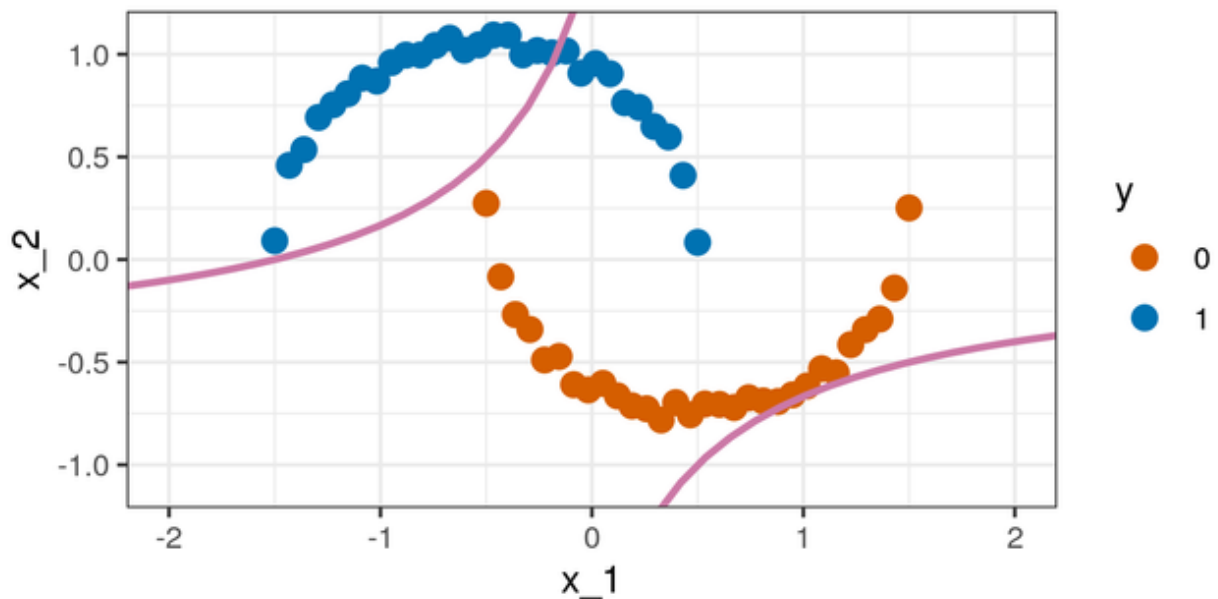
1 vote

Consider the binary classification problem in the figure. There are two predictors, x_1 and x_2 . Positive



4. classes (1) are shown in blue, negative classes (0) in orange. Which of the following methods generate the decision boundary in pink/purple?

0 correct answer
out of 0 respondent



Linear SVM

0%

0 votes

SVM with a
Gaussian kernel

0%

0 votes



SVM with a
polynomial kernel
of degree 2

0%

0 votes

LDA

0%

0 votes